



GSSS | GEETHA SHISHU
SHIKSHANA SANGHA(R)
**INSTITUTE OF ENGINEERING
AND TECHNOLOGY FOR WOMEN**
(Affiliated to Visvesvaraya Technological University, Belagavi, Approved by AICTE New Delhi & Govt of Karnataka)
KRS Road, Metagalli, Mysuru - 570016



8. Design and implement C/C++ Program to find a subset of a given set $S = \{s_1, s_2, \dots, s_n\}$ of n positive integers whose sum is equal to a given positive integer d .

```
#include<stdio.h>

void subset(int,int,int);

int x[10],w[10],d,count=0;

void main()
{
    int i,n,sum=0;

    printf("Enter the no. of elements: ");

    scanf("%d",&n);

    printf("\nEnter the elements in ascending order:\n");

    for(i=0;i<n;i++)

        scanf("%d",&w[i]);

    printf("\nEnter the sum: ");

    scanf("%d",&d);

    for(i=0;i<n;i++)

        sum=sum+w[i];

    if(sum<d)

    {

        printf("No solution\n");

        return;

    }

    subset(0,0,sum);

    if(count==0)

    {

        printf("No solution\n");
```



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```
return;

}

}

void subset(int cs,int k,int r)

{
int i;
x[k]=1;
if(cs+w[k]==d)
{
printf("\n\nSubset %d\n",++count);
for(i=0;i<=k;i++)
if(x[i]==1)
printf("%d\t",w[i]);
}
else
if(cs+w[k]+w[k+1]<=d)
subset(cs+w[k],k+1,r-w[k]);
if(cs+r-w[k]>=d && cs+w[k]<=d)
{
x[k]=0;
subset(cs,k+1,r-w[k]);
}
}
```

OUTPUT